



Neural State-Space models

Theory

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State-space models

They are defined in continuous/discrete time by:

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t); \theta) \\ y(t) &= g(x(t); \theta)\end{aligned}$$

$$\begin{aligned}x(k+1) &= f(x(k), u(k); \theta) \\ y(k) &= g(x(k); \theta)\end{aligned}$$

- x is the state vector. It is a latent (=hidden, unobserved) quantity
 - u is the input vector (=external/exogenous variable, covariate)
 - y is the output vector
 - θ is a vector of unknown parameters to be estimated

 - Powerful and flexible
 - Familiar to engineer/researcher
 - Suitable to embed physics
 - Might be harder to train than other sequence models
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Neural state-space models

In our context, f and g are feed-forward neural networks, e.g.

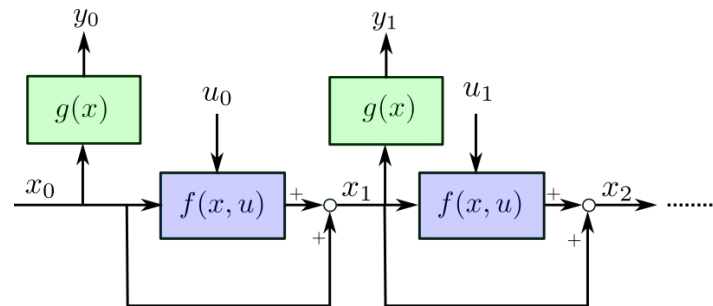
$$x(k+1) = x(k) + f(x(k), u(k); \theta) \quad f(z; \theta) = W_3 \sigma(W_2 \sigma(W_1 z + b_1) + b_2) + b_3$$

$$y(k) = g(x(k); \theta)$$

$$z = [x \ u]^T$$

$$\theta = \text{vec}(W_1, W_2, W_3, b_1, b_2, b_3)$$

Overall, we define a **neural state-space** model, aka **recurrent neural network (RNN)**



With respect to classic RNNs (Vanilla RNN, LSTM), focus on:

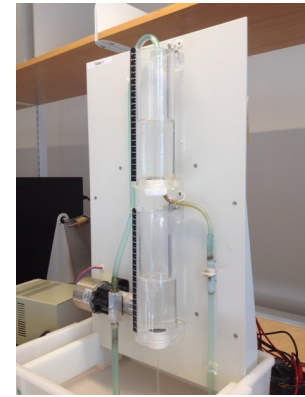
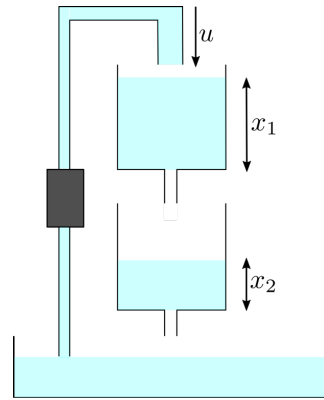
- Embedding prior (physical) knowledge
 - Parsimonious representations suitable for control
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Physics-inspired neural state-space models

Cascaded Tanks System. Input: upper tank inlet flow u . Output: lower tank level.

Intuitive physics:

- The system has two states: levels x_1, x_2
- The state x_2 is measured: $y = x_2$
- State x_1 does not depend on x_2
- State x_2 does not depend on u directly



Schoukens, M. et al. "Cascaded tanks benchmark combining soft and hard nonlinearities." Workshop on nonlinear system identification benchmarks, 2016.

Observations above are embedded in a **physics-inspired** neural state-space model:

$$\begin{aligned}\dot{x}_1 &= f_1(x_1, u; \theta) \\ \dot{x}_2 &= f_2(x_1, x_2; \theta) \\ y &= x_2\end{aligned}$$

Physics-inspired neural state-space models

Several contributions aim at embedding **prior knowledge** in the architecture.

- (Port) Hamiltonian neural networks

S. Greydanus et al., [Hamiltonian Neural Networks](#), 2019

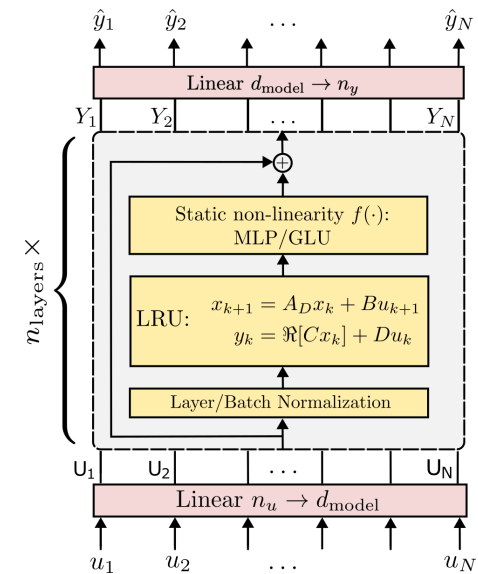
S. Moradi et al., [Port-Hamiltonian Neural Networks with Output Error Noise Models](#), 2025

- Lagrangian neural networks

M. Cranmer et al., [Lagrangian Neural Networks](#), 2020

- Structured state-space sequence models

A. Gu et al., [Efficiently Modeling Long Sequences with Structured State Spaces](#), 2022



From M. Forgione, M. Mejari, and D. Piga [Model order reduction of deep structured state-space models: A system-theoretic approach](#), 2024

... and many, many more!

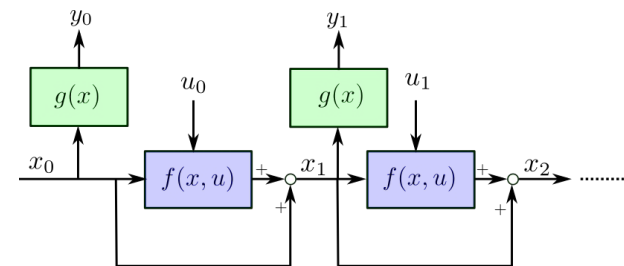
Simulation error minimization

In principle, **simulation error minimization** can be a valid approach:

$$\hat{\theta}, \hat{x}(0) = \arg \min_{\theta, x(0)} \frac{1}{T} \sum_{k=0}^{T-1} \|\hat{y}^{\text{sim}}(k) - y(k)\|^2$$

$$x(k+1) = x(k) + f(x(k), u(k); \theta)$$
$$\hat{y}^{\text{sim}}(k) = g(x(k); \theta)$$

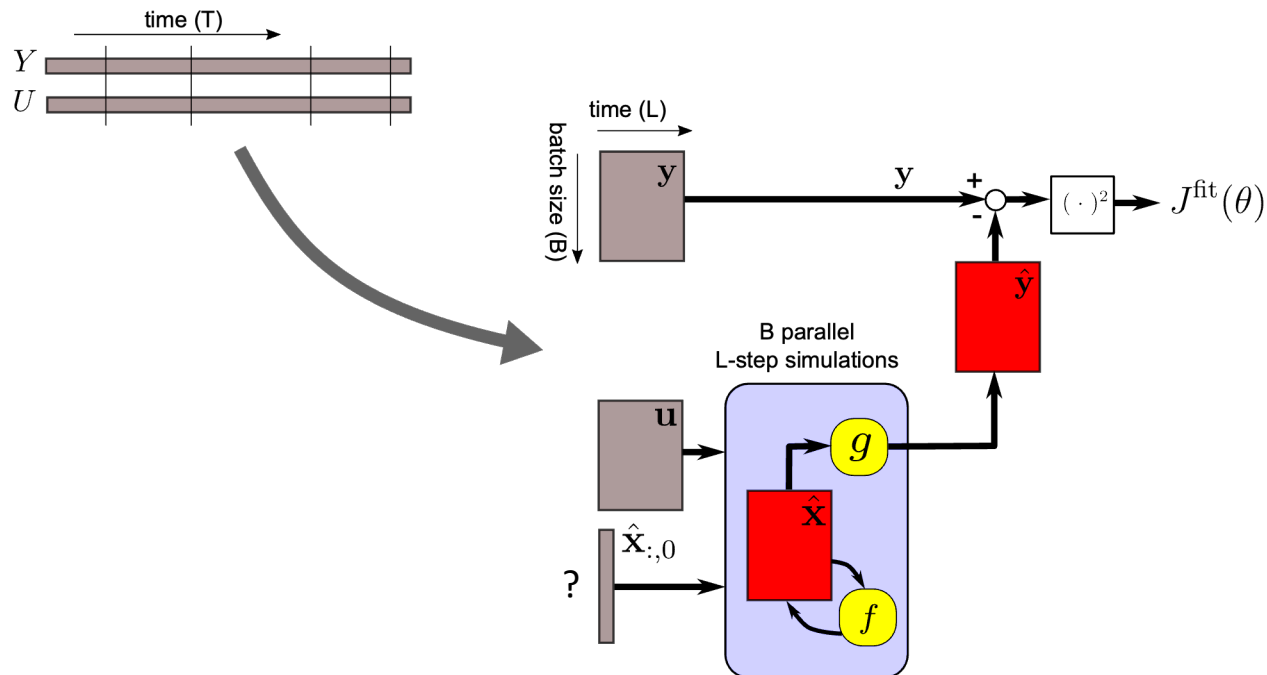
- Minimization wrt both θ (model) and x_0 (initial state)
- It is related to maximum likelihood under certain hypotheses
- It favors learning of **long-term** dependencies
- We stick with it for the exercise session
- It might be **computationally heavy**
 - Processing of a long sequence only gives one gradient update
- It might be **numerically hard**
 - Unstable/chaotic dynamics cannot be predicted on the long term
- It might not match the intended model usage
 - MPC only needs estimates up to the prediction horizon



For these reasons, alternative methods based on simulation on **shorter sub-sequences** have been developed

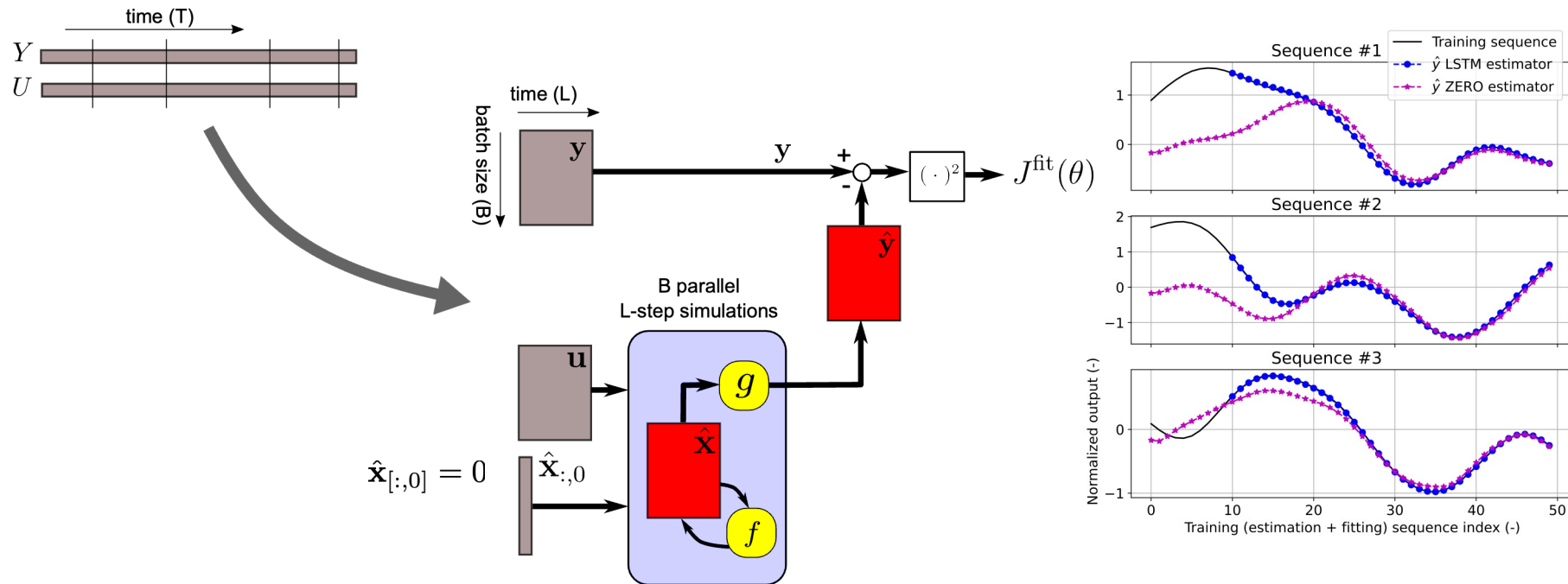
Truncated simulation error minimization

Different **L -step-ahead error minimization** schemes have been developed:



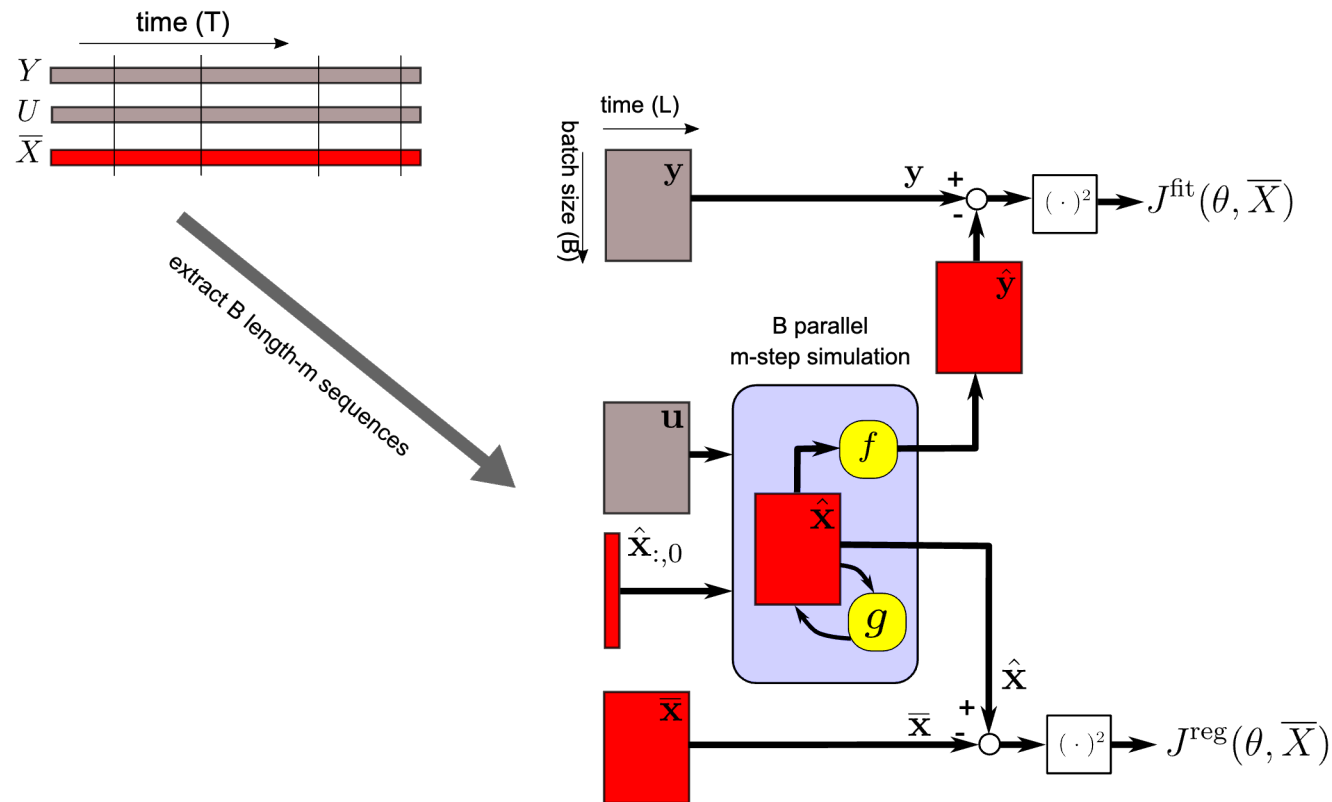
- Simulate B (batch size) shorter sequences of length L extracted from the longer (length T) training dataset.
- Main variation is: how do we deal with all possible initial conditions?

Zero initialization strategy



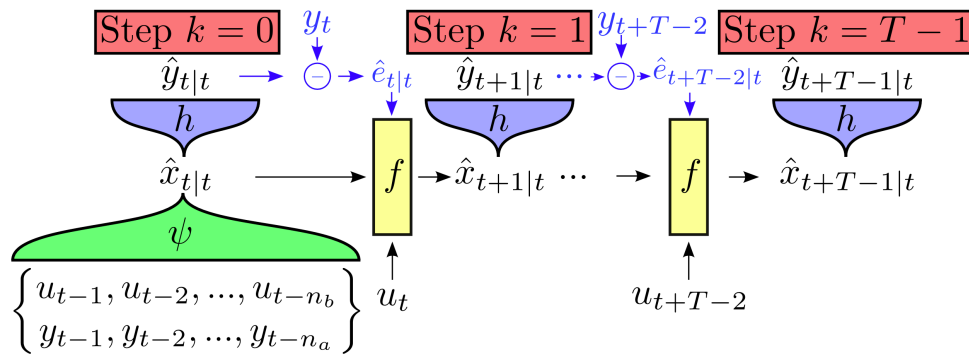
- Initialize from arbitrary (ZERO) state and discard a few initial samples from the loss
- Described as a baseline in M.Forgione, M.Mejari, and D.Piga, [Learning neural state-space models: do we need a state estimator?](#), 2022
- Very simple, yet effective when the system memory is short (wrt sequence length m)

Learn the hidden state sequence

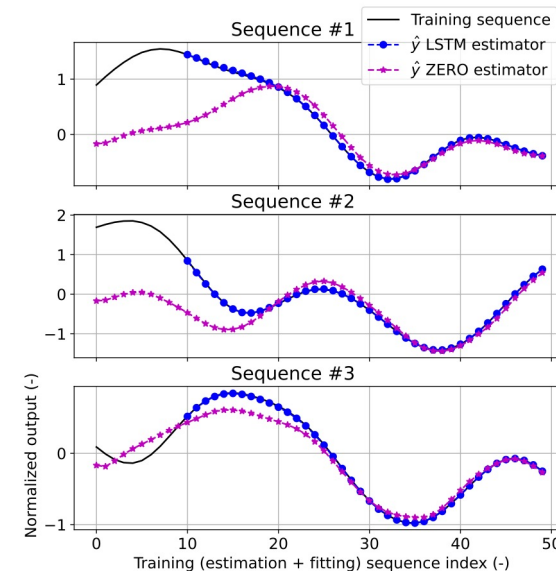


- Learn the sequence of states along with the system mappings f and g
- Regularize the learned state sequence with the system mapping
- Described in M. Forgione and D. Piga, [Continuous-time system identification with neural networks](#), 2021

Learn a state estimator along with the model (SUBNET)



From G. Beintema, M. Schoukens, R.Tóth [Deep subspace encoders for nonlinear system identification](#), 2022



- Use the initial samples of each sequence to reconstruct the initial state with a (learned) state estimator
- See G. Beintema, M. Schoukens, R.Tóth [Deep subspace encoders for nonlinear system identification](#), 2022

PyTorch Implementation

Unlike with standard RNNs, we must work at a slightly lower level

Custom state-update function $f(x, u)$

```
class NeuralStateUpdate(nn.Module):  
  
    def __init__(self, n_x=2, n_u=1, n_feat=32):  
        super(NeuralStateUpdate, self).__init__()  
  
        self.net = nn.Sequential(  
            nn.Linear(n_x+n_u, n_feat),  
            nn.Tanh(),  
            nn.Linear(n_feat, n_x),  
        )  
  
        for m in self.net.modules():  
            if isinstance(m, nn.Linear):  
                nn.init.normal_(m.weight, mean=0, std=1e-2)  
                nn.init.constant_(m.bias, val=0)  
  
    def forward(self, x, u):  
        z = torch.cat((x, u), dim=-1)  
        dx = self.net(z)  
        return dx
```

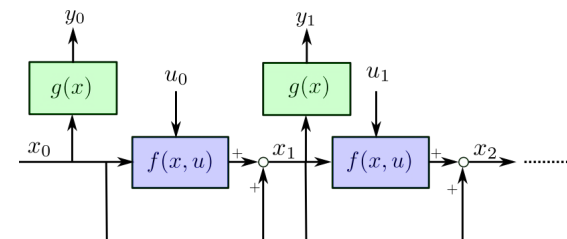
Custom code to unroll $f(x, u)$ over time

```
class StateSpaceSimulator(nn.Module):  
    def __init__(self, f_xu):  
        super().__init__()  
        self.f_xu = f_xu  
  
    def forward(self, x_0, u):  
        B, n_x = x_0.shape  
        _, T, _ = u.shape # B, T, n_u  
        x = torch.empty((B, T, n_x))  
        x_step = x_0  
  
        # manually unroll f_xu over time  
        for t in range(T):  
            x[:, t, :] = x_step  
            dx = self.f_xu(x_step, u[:, t, :])  
            x_step = x_step + dx  
  
        return x
```

Note:

- we actually implement: $x(k+1) = x(k) + f(x(k), u(k); \theta)$
- we make a custom weight initialization

Code adapted from <https://github.com/forgi86/pytorch-ident>



PyTorch Implementation

We can then instantiate a model that behaves very similarly to a standard PyTorch RNN

```
n_x = 2; n_u = 1;
f_xu = NeuralStateUpdate(n_x, n_u, n_feat=32)
simulator = StateSpaceSimulator(f_xu)
```

The model accepts a batch of initial states (B, n_x) and input sequences (B, T, n_u) and returns a batch of simulated states (B, T, n_x)

```
B, T = 32, 1024;
batch_x0 = torch.zeros((B, n_x))
batch_u = torch.randn((B, T, n_u)) # replace with actual training input
batch_x_sim = simulator(batch_x0, batch_u) # B, T, n_x
batch_x_sim.shape

torch.Size([32, 1024, 2])
```

We may finally process `batch_x_sim` through a feed-forward network to simulate the output equation $y = g(x)$. Alternatively, we may set the output to a subset of the simulated states.

Conclusions

- Neural state-space models are just custom RNNs
- Beyond standard Vanilla RNN, LSTM, GRU

Recent research has explored:

- State initialization for training with mini-batches
- Embedding of physical knowledge

We focus on physical knowledge in the exercises



Thank you for your attention